Structure-Compatible Model Selection Under Underspecification

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Abstract

Modern learning pipelines are often underspecified: many trained models can match train and in-distribution validation metrics while behaving differently under deployment shift. We study a finite, auditable slice of this problem. The central hypothesis is that, when a deployment shift is generated by a candidate transformation family, models should be selected not only by validation performance but also by whether their learned input-output map preserves that transformation. We consolidate six experimental phases spanning symbolic cyclic tasks, modular algorithmic tables, rotated visual objects, learned generators, language-like template substitution, frozen-encoder semantic retrieval, and OOD-free semantic model-zoo selection. Across these settings, structure-compatible predictors usually separate OOD behavior better than train accuracy, ID validation accuracy, loss, norm, or sharpness proxies. The strongest finished result is semantic model selection: among high train/ID candidates, learned compatibility selects mean OOD accuracy 0.978 versus 0.919 for train, ID, and random selectors, while wrong compatibility selects 0.751; a zoo-bootstrap gives learned-minus-random 0.059 with 95% CI [0.052, 0.065]. The claim is bounded: this is not universal OOD certification. It is a practical diagnostic and model-selection rule for finite settings where the relevant deployment structure is known, weakly inferred, or learned well enough to audit.

1 Introduction

Held-out validation can be silent about the distinction that matters most at deployment. A model may solve the training distribution by preserving a transportable rule, or by exploiting a shortcut that is equally accurate in-distribution but breaks under shift. This is the underspecification problem: standard pipelines can return many predictors with similar held-out performance and unstable deployment behavior [5].

This paper develops a direct model-selection response. When the expected deployment cases are generated by a transformation family, evaluate whether a candidate model preserves that transformation. The resulting score is not a generic regularizer for "more invariance." It is a structure-specific audit: the preserved structure must be the one that deployment will reuse, and wrong transformation families must fail as controls.

The empirical program asks three questions.

1. Can transformation compatibility predict OOD accuracy among models with similar train and ID evidence?
2. Can compatibility be weakened from oracle transformations to inferred or learned transformation families?
3. Can compatibility guide model selection or training without inspecting OOD labels?

The answer is positive in a bounded regime. Compatibility is strongest when the shift family is known or discoverable from overlap evidence; it weakens when the candidate family is ambiguous or when ID accuracy already tracks the shift. The important negative control is equally informative: wrong compatibility can be null or anti-predictive, so the method is not merely rewarding generic smoothness or invariance.

Contributions. First, we give a unified formulation of structure-compatible model selection under underspecification. Second, we consolidate six result phases into one evidence ladder from oracle finite transformations to learned semantic selection. Third, we report the strongest deployment-like selector: learned semantic compatibility improves OOD-free model selection over train/ID/random selectors and fails under wrong compatibility. Fourth, we connect the same methodological rule to a companion finite-agent benchmark appendix: behavior alone is insufficient; a pass requires a structure-specific gate and anti-cheat controls.

2 Problem Setup

Let $f : \mathcal{X} \to \mathcal{Y}$ be a trained predictor selected from a model zoo $\mathcal{F}$. Let D_{tr} and D_{id} be training and in-distribution validation distributions, and let D_{ood} be generated from the same task by a deployment-relevant transformation family $\mathcal{G}$. In underspecified regimes, many $f \in \mathcal{F}$ satisfy similar $A_{tr}(f)$ and $A_{id}(f)$ while varying in $A_{ood}(f)$.

For each transformation $g \in \mathcal{G}$, let τ_g be the corresponding label transport when labels transform. A model is structure-compatible when

$$f(gx) \approx \tau_g(f(x)) \tag{1}$$

on audited examples. In classification tasks we use agreement rates; in retrieval or semantic tasks we use transportconsistent nearest-neighbor or score agreement. The compatibility score is

$$C_{\mathcal{G}}(f) = \mathbb{E}_{x,g}\left[s\left(f(gx), \tau_g(f(x))\right)\right], \tag{2}$$

where s is the task-appropriate agreement score.

Three variants appear in the evidence ladder:

- **Oracle compatibility** uses the true deployment-generating transformation family.
- **Inferred or discovered compatibility** constructs a supported family from train-label overlap, candidate transports, or frozen-encoder neighborhoods.
- **Wrong compatibility** uses irrelevant transformations as a negative control.

For OOD-free model selection, the selector chooses

$$\hat{f} = \arg \max_{f \in \mathcal{F}_{eligible}} C_{\hat{\mathcal{G}}}(f), \tag{3}$$

where $\mathcal{F}_{eligible}$ is restricted to high train/ID candidates and $\hat{\mathcal{G}}$ is discovered without OOD labels. OOD labels are then used only for evaluation.

Discovery procedure. In the finite modular setting, discovered transformations are admitted only when observed train-label overlaps support them. A shift g is admitted if at least m observed examples have their transformed counterpart observed and all such overlaps satisfy the expected label transport. If no nontrivial shift is supported, identity remains as the defined lower case. Later phases replace this finite shift rule with learned candidate transports, template substitutions, or frozen-encoder neighborhoods, but preserve the same audit pattern: score candidate transforms without OOD labels, then evaluate selected models on held out OOD data.

Tie handling and uncertainty. When a selector ties among eligible candidates, the current semantic-selection summary reports the mean OOD of tied candidates rather than choosing a favorable tie-break. This is conservative with respect to cherry-picking. For the Phase 6 selector, we regenerate the finite payload and report 1,000-rep bootstrap intervals clustered by selection zoo. Earlier phases still rely on archived point summaries unless their raw row payloads are restored or regenerated.

3 Experimental Ladder

The suite spans six phases. Table 1 summarizes the role of each phase. The point of the ladder is not to claim that each phase is equally realistic. It is to make each relaxation explicit: from true transformations, to supported inference, to learned generators, to finite text, to frozen semantic encoders, to OOD-free selection.

Baselines and controls. The selector baselines are train accuracy, ID validation accuracy, random eligible candidate, loss, parameter norm, and sharpness where available. The primary anti-cheat control is compatibility with a wrong transformation family. When regularization or augmentation is used, we compare against unregularized, oracle, learned, random, or wrong-substitution variants as appropriate.

4 Results

4.1 Compatibility predicts OOD in finite domains

In the initial L4 suite, true compatibility is the top predictor for symbolic cyclic, modular neural, and vision rotation domains; inferred compatibility is perfect in the exact modular sanity check and near true compatibility in the symbolic domain. Selection by true compatibility reaches mean selected OOD accuracy 1.000 across the three domains where the selector is available, compared with 0.660 for train or ID accuracy and 0.458 for wrong compatibility.

4.2 Discovered compatibility can control OOD

Phase 2 replaces a purely oracle post-hoc score with supported modular shifts inferred from observed train-label overlaps. Compatibility regularization improves mean OOD from 0.134 with no regularization to 0.320 at weight 0.05 and 0.352 at weight 0.20. In the high-ID subset, OOD improves from 0.178 to 0.519 and 0.573 respectively.

Phase 3 learns candidate generators. In modular tasks, discovered compatibility correlates with OOD at Pearson $r = 0.787$, while wrong compatibility is negative. Regularization improves high-ID OOD from 0.256 to 0.425 and 0.502. In a vision transfer arm, learned augmentation improves paired OOD by 0.391 over no augmentation, near the oracle augmentation delta of 0.440.

4.3 Finite language and semantic transfer

The finite text bridge renders modular examples as language-like templates and learns number-word/template substitutions. Discovered compatibility predicts OOD at $r = 0.858$, while ID and train accuracy are 0.478 and wrong compatibility is -0.088. Compatibility regularization improves high-ID OOD from 0.224 to 0.512. Wrong substitution augmentation is a strong negative control: mean OOD 0.076 versus 0.421 for partial correct substitution.

The semantic retrieval phase uses frozen encoders rather than supplied symbolic substitutions. With two encoders, true compatibility predicts OOD at $r = 0.940$ and discovered compatibility at $r = 0.861$; wrong compatibility is anti-predictive at $r = -0.872$.

Phase	Setting	Main operation	Claim boundary
1	symbolic, modular, vision	oracle and inferred compatibility	finite known transformations
2	modular neural	discovered shifts plus regularization	finite structured intervention
3	modular and vision	learned candidate generators	finite generator transfer
4	language templates	learned substitutions	finite text-like substitution
5	frozen semantic retrieval	encoder-neighborhood transports	bounded semantic retrieval
6	semantic model zoos	OOD-free selector among high-ID candidates	bounded semantic model selection

Table 1: Evidence ladder. Each phase relaxes the transformation source while preserving negative controls and OOD-held-out evaluation.

Domain	top compatible r	ID r	N
modular neural	0.649	0.188	128
symbolic cyclic	0.763	0.099	128
vision rotation	0.451	0.445	96

Table 2: Phase 1 Pearson correlations with OOD accuracy. Compatibility most clearly separates symbolic and modular settings; vision is a closer case where ID accuracy is also informative.

Selector	selected OOD	regret
random candidate	0.919	0.073
ID validation	0.919	0.073
train accuracy	0.919	0.073
wrong compatibility	0.751	0.242
learned compatibility	0.978	0.014
true compatibility	0.993	0.000
OOD oracle	0.993	0.000

Table 3: Phase 6 semantic model selection. Learned compatibility separates high-ID candidates before OOD labels are inspected.

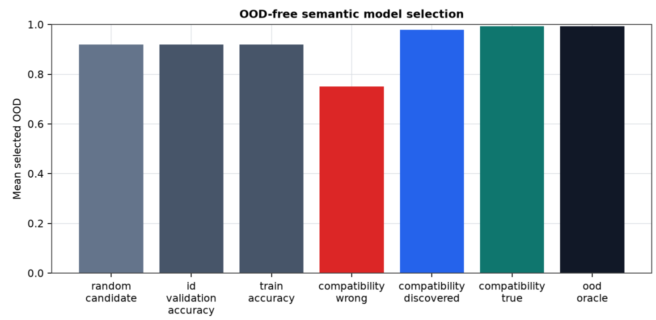

Figure 1: Semantic selector performance. The main paper uses this as the headline visual because it is the closest deployment analogue: choose among ID-equivalent candidates before seeing OOD labels.

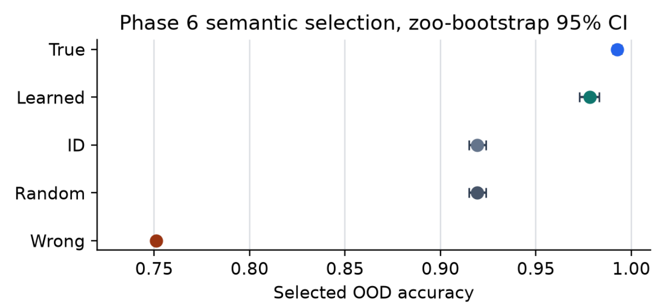

Figure 2: Zoo-bootstrap intervals for Phase 6 selector performance. Learned compatibility separates from train/ID/random and from the wrong-compatibility control under the model-zoo bootstrap unit.

4.4 OOD-free semantic model selection

The strongest finished result is the selector test. Across 120 eligible semantic model zoos, train accuracy, ID validation, and random candidate selection all select mean OOD 0.919. Learned compatibility selects 0.978 with regret 0.014, while wrong compatibility selects 0.751. A 1,000-rep selection-zoo bootstrap gives learned selected OOD 0.978 [0.973, 0.983], learned-minus-random 0.059 [0.052, 0.065], learned-minus-ID 0.059 [0.052, 0.065], learned-minus-wrong 0.227 [0.221, 0.233], and learned regret 0.014 [0.011, 0.018]. True compatibility and the OOD oracle select 0.993.

5 Relation to Causally Grounded Agents

The companion benchmark work uses the same rule in an agent setting. A finite agent is not counted as grounded merely because it succeeds; it must pass a behavior gate, a structure-specific gate, and anti-cheat controls. For example, Suite C tests whether an inquiry policy reopens after world change, recovers attribution, avoids false calm, spends fewer probes than high-cost controls, and reopens after a second shift. Suite D/E tests whether a future-critical variable survives memory, generated action, tool schema, repair, and causal readout surfaces. This paper does not claim those suites as evidence for broad agency. They are included because they show the same evaluation pattern beyond static model selection: success should be accepted only when the right structure did the work.

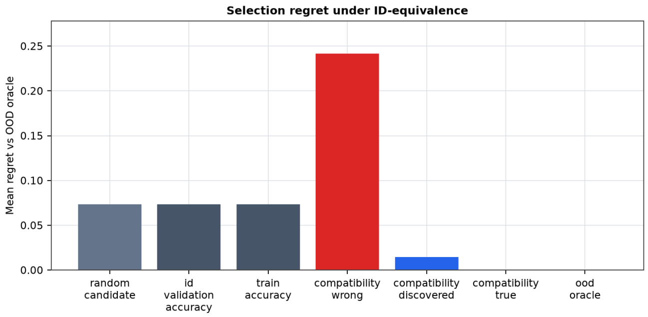

Figure 3: Semantic selector regret. Learned compatibility approaches the true compatibility/OOD-oracle ceiling and sharply separates from wrong compatibility.

6 Related Work

Underspecification and shortcut learning. The motivation follows work showing that modern ML pipelines can be underspecified and deployment-unstable despite strong held-out performance [5]. Shortcut learning makes the same concern concrete: models may exploit decision rules that solve benchmark distributions without preserving the intended structure [6].

Invariance, equivariance, and OOD generalization. Invariant risk minimization and related domain-generalization methods aim to learn stable predictors across environments [3]. Our setting differs in emphasis: we audit and select among already trained finite models using a candidate deployment transformation family, and we require wrong transformation controls so that "more invariance" is not treated as a generic good.

Causal and disentangled representations. Causal representation learning argues that transfer and generalization benefit from recovering variables and mechanisms that match the data-generating process [4, 11]. Disentanglement work cautions that unsupervised factor recovery requires inductive biases and cannot be assumed from reconstruction alone [9]. Our method is compatible with that caution: it uses explicit or discovered candidate transforms and evaluates them with negative controls.

Probing and interpretability controls. Probe accuracy alone can reflect probe capacity rather than represented structure [7]. The same logic motivates our compatibility controls: the score matters when wrong transformations fail and when the selected model improves held-out OOD behavior. Causal intervention work in language models likewise motivates asking where a representation is load-bearing rather than merely decodable [10].

Agent evaluation. Language-agent systems such as ReAct, Reflexion, Voyager, SayCan, and Tree of Thoughts show the value of interleaving reasoning, acting, feedback, skills, and search [1, 12–15]. Safety work on reward hacking, distribution shift, and goal misgeneralization argues that behavior-level success can hide proxy failure [2, 8]. The companion benchmark appendix imports that lesson into finite diagnostic suites with explicit structure gates.

7 Limitations

The current claim is intentionally finite. It does not solve OOD generalization, certify open-world semantics, or prove that transformation discovery is reliable in arbitrary domains. Several results still use oracle or partly supported transformation families. Some domains, especially vision, contain cases where ID accuracy remains competitive with compatibility. The semantic results use frozen encoders and constructed retrieval/model-zoo settings, not unconstrained natural-language deployment. Finally, the agent-benchmark connection is methodological; it does not establish consciousness, production reliability, or general autonomy.

8 Conclusion

Structure-compatible model selection gives a practical answer to one slice of underspecification. When validation cannot distinguish shortcut and transportable solutions, ask whether the learned function preserves the transformations deployment will require. The result is not a universal certificate, but it is a useful audit surface: wrong transformations can fail, regularization can improve OOD without OOD labels, and learned compatibility can select better semantic retrieval models among ID-equivalent candidates. The broader lesson is shared with proxy-resistant agent benchmarks: behavior only counts when the right structure did the work.

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A Detailed Result Ledger

This appendix records the source-backed result tables used by the main paper. All paths are relative to the repository root.

A.1 Phase 1: Cross-Domain Structure Compatibility

Source: `experiments/structure_compatible_generalization/results/structure_compatible_l4_2026_07_06.md`.

Domain	rows	mean train	mean ID	mean OOD
modular exact	2	1.000	1.000	0.545
modular neural	128	0.732	0.732	0.165
symbolic cyclic	128	0.973	0.984	0.394
vision rotation	96	0.773	0.773	0.380

Table 4: Phase 1 domain sizes and mean accuracies.

Selector	domains	mean selected OOD	note
true compatibility	3	1.000	oracle structure
ID validation accuracy	3	0.660	standard selector
train accuracy	3	0.660	standard selector
inferred compatibility	2	0.531	unavailable in one domain
wrong compatibility	3	0.458	negative control
parameter L2	3	0.426	proxy baseline
sharpness proxy	3	0.410	proxy baseline
negative train loss	3	0.117	proxy baseline

Table 5: Phase 1 OOD-free selection by predictor family.

A.2 Phase 2: Inferred Transformations and Intervention

Source: `experiments/structure_compatible_generalization/results/phase2_transformations_2026_07_06.md`.

The main regime transition is from post-hoc oracle compatibility to supported modular shifts inferred from observed train-label overlaps. The evidence-bearing result is regularization, not the one-domain selector sanity check.

reg.	N	mean train	mean OOD	high-ID N	high-ID OOD
0.000	180	0.719	0.134	98	0.178
0.050	180	0.719	0.320	98	0.519
0.200	180	0.723	0.352	99	0.573

Table 6: Phase 2 compatibility regularization improves OOD while preserving a high-ID comparison surface.

A.3 Phase 3: Learned Generators and Vision Transfer

Source: `experiments/structure_compatible_generalization/results/phase3_learned_generators_2026_07_06.md`.

A.4 Phase 4: Language-Like Template Substitution

Source: `experiments/structure_compatible_generalization/results/language_template_substitution_2026_07_06.md`.

Domain	predictor	Pearson r	Spearman r	N
modular learned generator	true compatibility	0.812	0.367	270
modular learned generator	discovered compatibility	0.787	0.474	270
modular learned generator	ID validation	0.398	0.120	270
modular learned generator	wrong compatibility	-0.085	0.049	270
vision learned generator	true compatibility	0.678	0.591	144
vision learned generator	discovered compatibility	0.599	0.502	144
vision learned generator	ID validation	0.661	0.632	144

Table 7: Phase 3 predictor correlations. The vision arm is a mixed case where ID remains strong, so the paper should not overstate compatibility dominance.

Vision regime	N	mean OOD	mean learned compatibility
none	36	0.258	0.486
learned augmentation	36	0.644	0.793
oracle augmentation	36	0.693	0.801
random augmentation	36	0.610	0.760

Table 8: Phase 3 vision transfer arm. The random augmentation result means the vision claim should be framed as transfer evidence with a nontrivial control, not as a pure learned-generator victory.

A.5 Phase 5: Semantic Retrieval Transfer

Source: `experiments/structure_compatible_generalization/results/semantic_retrieval_transfer_2026_07_06.md`.

A.6 Phase 6: Semantic Selection Control

Source: `experiments/structure_compatible_generalization/results/semantic_selection_control_2026_07_06.md`. Bootstrap source: `experiments/structure_compatible_generalization/results/semantic_selection_bootstrap_2026_07_06.md`.

All registered selector gates passed: minimum zoo count, beats ID validation, beats train accuracy, beats random candidate, wrong control fails, and accepted. The table in the main paper reports the full selector comparison. The regenerated 120-zoo payload gives learned-minus-random selected OOD 0.059 with 95% CI [0.052, 0.065], learned-minus-ID 0.059 [0.052, 0.065], learned-minus-wrong 0.227 [0.221, 0.233], and learned regret 0.014 [0.011, 0.018].

B Companion Benchmark Appendix

The companion benchmark is not the main ICML claim, but it is useful because it shows the same anti-proxy rule in finite agents.

B.1 Minimum Pass Rule

Source: `docs/causally_grounded_agents_benchmark.md`.

A suite does not pass on behavior alone. A pass requires:

1. the task behavior gate,
2. the relevant causal-structure gate, and
3. anti-cheat controls specific to that suite.

The score should remain vector-valued: behavior, causal representation, attribution, inquiry, commitment, and generalization.

Predictor	Pearson r	Spearman r	N
discovered compatibility	0.858	0.596	256
true compatibility	0.853	0.628	256
negative train loss	0.492	0.232	256
ID validation	0.478	0.127	256
train accuracy	0.478	0.127	256
wrong compatibility	-0.088	0.059	256
sharpness proxy	-0.111	0.063	256
parameter L2	-0.476	-0.189	256

Table 9: Phase 4 finite-text bridge.

reg.	N	train	ID	OOD	high-ID N	high-ID OOD
0.000	64	0.554	0.554	0.147	29	0.224
0.050	64	0.553	0.553	0.234	29	0.415
0.200	64	0.555	0.555	0.255	29	0.463
0.500	64	0.558	0.558	0.277	29	0.512

Table 10: Phase 4 compatibility regularization.

B.2 Suite C: Re-Engagement Under World Change

Sources: `experiments/world_responds/results/suite_c_reengagement_2026_07_06.md` and `experiments/world_responds/results/suite_c_neural_transfer_2026_07_06.md`.

The hand-policy headline condition is `burst_then_refractory`. It passes C1-C6 with final affected MAE 0.112, first-shift selectivity 16.927, second-shift reopenability 16.667, and 22.6 probes. The signal-layer `fixed_surprise_decrement` control fails correctly: it looks quiet but ends at final MAE 0.517.

The learned probe head passes the same held-out C1-C6 gate with final affected MAE 0.112, first-shift selectivity 16.667, second-shift reopenability 17.448, and 23.1 probes. Stale-signal, wrong-signal, signal-suppression, and matched-random controls fail in the intended way.

B.3 Suite D/E: Long-Horizon Commitment Surfaces

Source: `experiments/long_horizon_bottleneck/BENCHMARK_CARD.md`.

The long-horizon benchmark asks whether a future-critical variable survives the surfaces where it must control action: hidden state, generated JSON, tool arguments, repair branches, and value-token causal readouts. The strongest lesson is the commitment-surface law: memory becomes agent-relevant when it is bound to a future action or tool surface.

C Reproducibility Notes

The source reports list base seeds, model/config counts, GPU type, and budget checks. The paper package should be submitted with:

- the six source result reports above,
- figure regeneration scripts under `scripts/build_structure_compatible_*.py` and `scripts/export_structure_compatible_artifacts.py`,
- task code under `experiments/structure_compatible_generalization/`,
- public benchmark schema under `docs/causally_grounded_agents_release_schema.json`.

D Reviewer-Facing Weaknesses

The strongest reviewer objections are:

1. Some phases use oracle or partly hand-designed transformation families.

Predictor	Pearson r	Spearman r	N
true compatibility	0.940	0.915	64
discovered compatibility	0.861	0.863	64
train accuracy	0.772	0.628	64
ID validation	0.709	0.488	64
wrong compatibility	-0.872	-0.919	64

Table 11: Phase 5 frozen-encoder semantic retrieval.

2. The vision arm has a strong ID baseline and a strong random augmentation control.
3. The semantic tasks are constructed around frozen encoders and model zoos; they are not broad natural-language deployment.
4. Phase 6 now has bootstrap intervals, but earlier phases still need row-level uncertainty if they keep quantitative table space in the main paper.
5. The finite-agent benchmark material should remain an appendix unless a separate benchmark paper is submitted.

E Next Experiments

The next ICML-strengthening experiments are:

- deterministic tie-break stress tests for the Phase 6 selector lift and wrong-control gap,
- larger semantic model zoos with a third frozen encoder,
- held-out transformation-family discovery where candidate transforms are learned from one family and evaluated on another,
- an ablation that holds augmentation volume fixed while varying transformation correctness,
- less privileged source-estimation and open-agent transfer for the teacher-free Suite C companion benchmark.